

Performance Testing

Date	15 March 2026
Team ID	xxxxxx
Project Name	Rising Water
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	e.g., JMeter, Locust, k6, Postman
Type of Testing	e.g., Load Testing, Stress Testing, Spike Testing
Target Module / API	Specify which feature or endpoint was tested
Test Environment	e.g., Local / Cloud / Hosted Server
Test Date	

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Load Test - Fetch current water level data from API	200	300	Response time < 1.5s, 0% error rate
2	Stress Test - Multiple sensors push rising water data simultaneously	1000	600	System handles data without crashing, alerts triggered
3	Spike Test - Sudden spike: water level crosses danger threshold	50 to 800	180	Alert notification sent within 5s to all users
4	Endurance Test - Continuous monitoring of water level for 24h	100	86400	No memory leak, data logging remains stable

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 seconds	Pass	Stable under load
2	Response Time (Max)	< 5 seconds	3.8 seconds	Pass	1 spike during sensor burst
3	Throughput (Req/sec)	150	168	Pass	System handled more than target
4	Error Rate	< 1%	0.3%	Pass	Few timeout errors at peak
5	CPU Utilization	< 80%	72%	Pass	CPU peaked at 8th minute
6	Memory Utilization	< 80%	65%	Pass	No memory leak observed

Step 4: Observations & Analysis

Key Findings:
 [Describe what the test results revealed about your system's performance]

Bottlenecks Identified:
 [List any performance bottlenecks observed during testing]

Optimization Steps Taken:
 [Describe any changes or optimizations made to improve performance]

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

